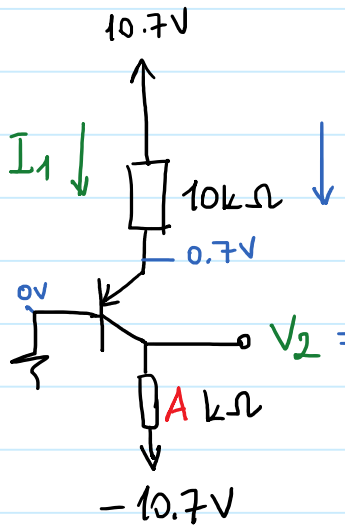
Q2

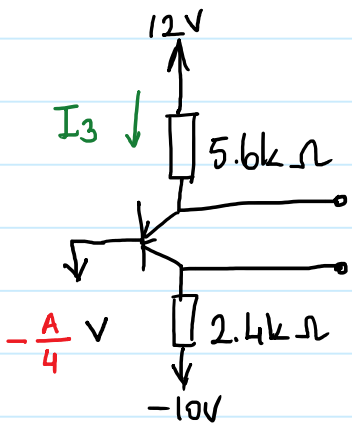
Assume β is very large. Find the unknowns.
(The parameter A is calculated in Q1. Round your result to the nearest integers.)



$\frac{10.7-0.7}{10k} = 1mA = 1000\,\mu A$

$\beta \to \infty$, $\alpha \to 1$

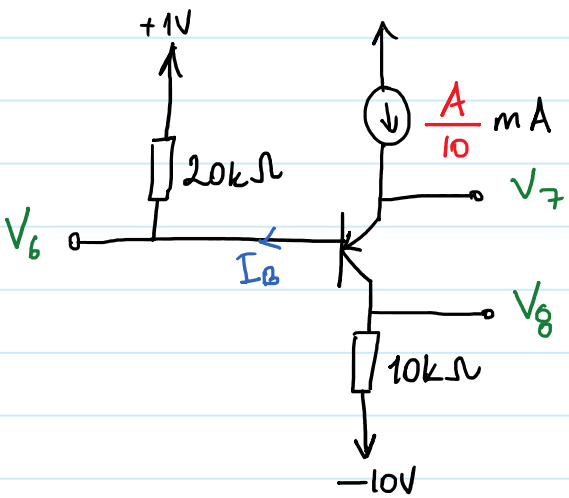
$V_2 = -10.7 + A\ V = (-10.7+A)\times 1000\ mV$



$I_3 = \frac{12-V_4}{5.6k} \times 10^6\ \mu A$

$V_4 = -A/4 + 0.7\ V = (-A/4+0.7)\times 1000\ mV$

$V_5 = (-10 + I_3 \times 2.4k)\times 1000\ mV$



$\beta \to \infty$, $I_B \to 0\ A$

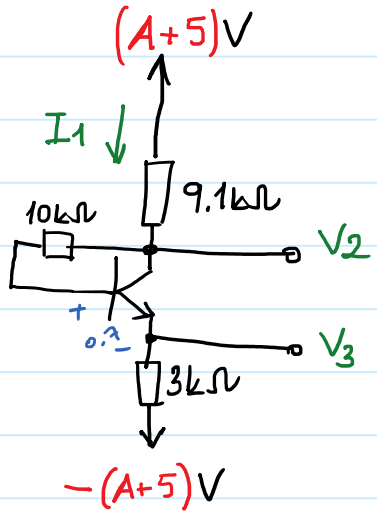
$V_6 = 1 + 20k.I_B = 1000\ mV$

$V_7 = V_6 + 0.7\ V = 1700\ mV$

$V_8 = -10 + 10\frac{A}{10} = (A-10)\times 1000\ mV$

Q3

Assume β is very large. Find the unknowns.
(The parameter A is calculated in Q1. Round your result to the nearest integer.)



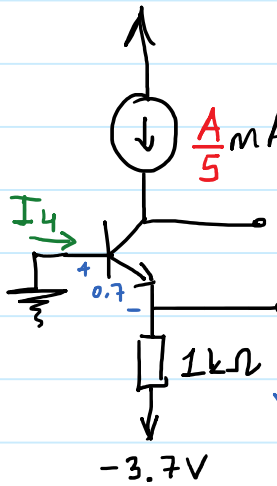
$\beta \to \infty$, $I_B \to 0$ A

$2A+10 = I_1(9.1k) + \overset{0}{I_B(10k)} + 0.7 + I_1(3k)$

$I_1 = \frac{2A+9.3}{12.1} \times 1000\ \mu A$

$V_2 = \left[A+5 - I_1(9.1k)\right] \times 1000\ mV$

$V_3 = \left[-A-5 + I_1(3k)\right] \times 1000\ mV$



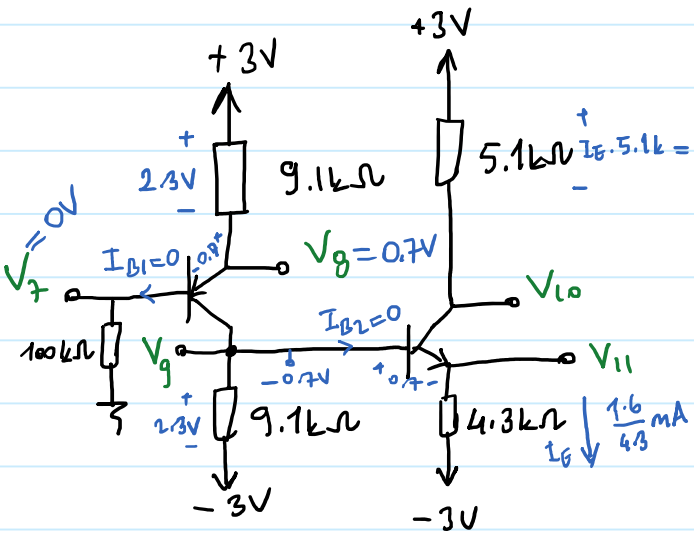
$\frac{A}{5}$ mA < 2 mA $\quad (1 < A < 10)$

$V_5 = -0.7 + 0.2 = -0.5V = -500\ mV$ (saturation)

$V_6 = -0.7\ V = -700\ mV$ $\quad$ [saturation $\rightarrow I_4 \neq 0$]

$I_E = \frac{-0.7+3.7}{1}\ mA = 3000\ \mu A$

$I_4 = I_E - A/5\ m = (3 - A/5) \times 1000\ \mu A$



$\beta \to 0$

$V_7 = 0\ mV$

$V_8 = 700\ mV$

$V_9 = -3 + 2.3 = -0.7V = -700\ mV$

$V_8 - V_9 > 0.3$, active mode ✓

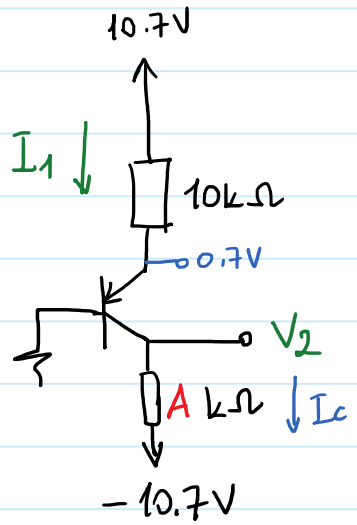
$V_{11} = -0.7 - 0.7 = -1.4\ V = -1400\ mV$

$V_{10} = 3 - 1.9 = 1.1V = 1100 mV$

Q4

Assuming β=100, repeat Q2

(The parameter A is calculated in Q1. Round your result to the nearest integer.)

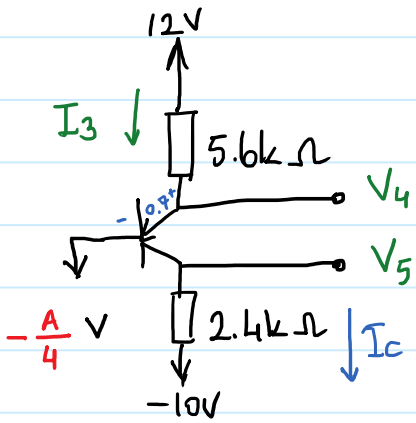


$\beta = 100 \rightarrow \alpha = \frac{\beta}{\beta+1} = 0.99$

$I_1 = \frac{10.7-0.7}{10k} = 1mA = 1000\mu A$

$I_C = \alpha I_1 = 0.99mA = 990\mu A$

$V_2 = \left(-10.7 + (0.99)A\right) \times 1000 \ mV$

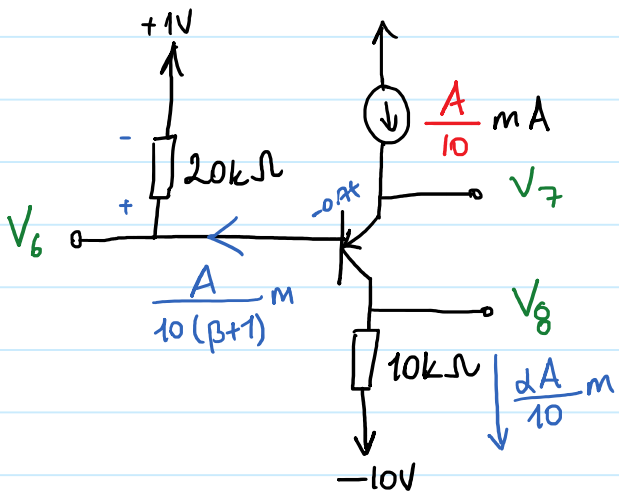


$V_4 = \left(-A/4 + 0.7\right) \times 1000 \ mV$

$I_3 = \left[(12 - V_4)/5.6\right] \times 1000 \ \mu A$

$I_C = \alpha I_3 = 0.99 I_3$

$V_5 = \left(-10 + 2.4k \cdot \alpha I_3\right) \times 1000 \ mV$



$V_6 = 1 + \frac{20A}{10(\beta+1)} V = \left(1 + 2A/_{\beta+1}\right) \times 1000 \ mV$

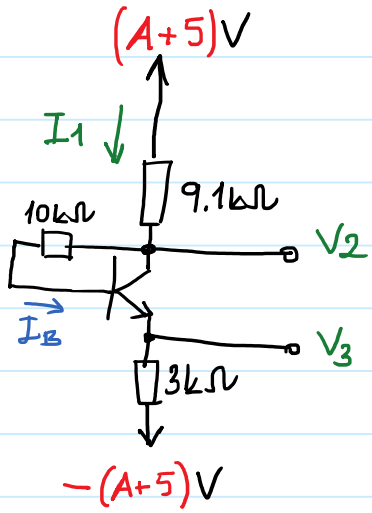
$V_7 = \left(V_6 + 0.7\right) \times 1000 \ mV$

$V_8 = -10 + 10 \cdot dA/10 \ V = \left(-10 + dA\right)_{\times 1000 \ mV}$

Q5

Assuming β=100, repeat Q3

(The parameter A is calculated in Q1. Round your result to the nearest integer.)



$\beta = 100$, $\alpha = 0.99$

$2A+10 = \beta I_B(9.1k) + I_B(10k) + 0.7 + (\beta+1) I_B\, 3k$

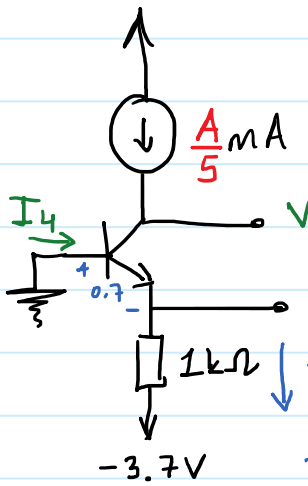
$2A+9.3 = I_B\, 1223k$

$I_B = (2A+9.3)/1223$ mA

$I_1 = \beta I_B = \left(100/1223 \times (2A+9.3)\right) \times 1000$ μA

$V_2 = (A+5-(9.1k)I_1) \times 1000$ mV

$V_3 = \left(-A-5+(3k).101.I_B\right) \times 1000$ mV



$\frac{A}{5}$ mA < 2 mA $\quad (1 < A < 10)$

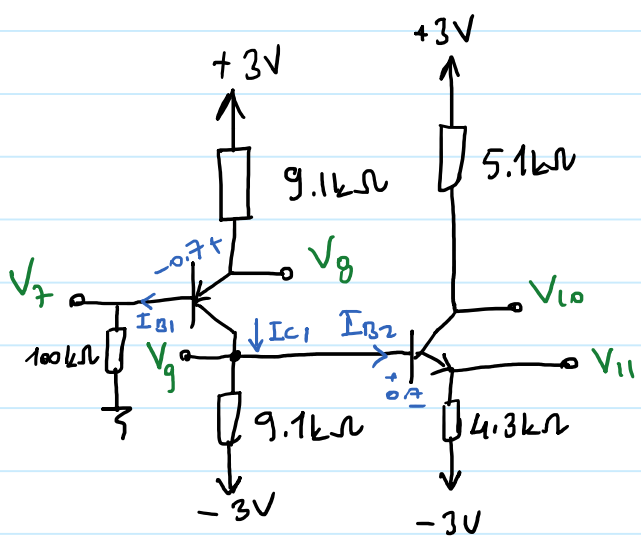
$V_5 = -0.7 + 0.2 = -0.5\,V = -500$ mV (saturation)

$V_6 = -0.7\,V = -700$ mV

[saturation → $I_4 \neq 0$]

$I_E = \frac{-0.7+3.7}{1}$ mA $= 3000$ μA

$I_4 = I_E - A/5\,m = (3 - A/5) \times 1000$ μA



Assume active mode op.

$3 = 9.1k(\beta+1) I_{B1} + 0.7 + I_{B1}\, 100k$

$2.3 = 1019k.I_{B1} \Rightarrow I_{B1} = 2.26$ μA

$V_7 = 100k\, I_{B1} = 226$ mV

$V_8 = V_7 + 0.7 = 926$ mV

$I_{C1} = I_{B2} + \frac{V_9+3}{9.1k} = \beta I_{B1} = 226$ μA

$V_9 = 0.7 + (\beta+1) I_{B2}(4.3k) - 3$

$V_9 = -2.3 + 4343k\left(226\mu - \frac{V_9+3}{9.1k}\right) = -946$ mV

$V_{11} = V_9 - 0.7 = -1646$ mV $\quad V_{10} = 1410$ mV